

Лабораторная №7: Repository/JSON

Задание

1. Ознакомьтесь со структурой и содержимым JSON-файла с данными (имя, фамилия, путь к jpg-файлу фотографии) известных людей (celebrities) в области IT.

```
[
  {
    "Id": 1,
    "Firstname": "Noam",
    "Surname": "Chomsky",
    "PhotoPath": "/Photo/Chomsky.jpg"
  },
  {
    "Id": 2,
    "Firstname": "Tim",
    "Surname": "Berners-Lee",
    "PhotoPath": "/Photo/Berners-Lee.jpg"
  },
  {
    "Id": 3,
    "Firstname": "Edgar",
    "Surname": "Codd",
    "PhotoPath": "/Photo/Codd.jpg"
  },
  {
    "Id": 4,
    "Firstname": "Donald",
    "Surname": "Knuth",
    "PhotoPath": "/Photo/Knuth.jpg"
  },
  {
    "Id": 5,
    "Firstname": "Linus",
    "Surname": "Torvalds",
    "PhotoPath": "/Photo/Torvalds.jpg"
  },
  {
    "Id": 6,
    "Firstname": "John",
    "Surname": "Neumann",
    "PhotoPath": "/Photo/Neumann.jpg"
  },
  {
    "Id": 7,
    "Firstname": "Edsger",
    "Surname": "Dijkstra",
    "PhotoPath": "/Photo/Dijkstra.jpg"
  },
  {
    "Id": 8,
    "Firstname": "Marvin",
    "Surname": "Minsky",
    "PhotoPath": "/Photo/Minsky.jpg"
  }
]
```

2. Исследуйте интерфейс **ICelebrity**, описывающий контракт между приложением **DP007_json_test** и библиотекой **DP007_json.lib** реализующей интерфейс **ICelebrity**.

```
public interface ICelebrity<T> : IDisposable
{
    List<T> GetAllCelebrities();           // получить все Знаменитости
    T? GetCelebrityById(int Id);           // получить Знаменитость по Id
    bool DelCelebrity(int id);              // удалить Знаменитость по Id
    bool AddCelebrity(T celebrity);         // добавить Знаменитость
    int AddCelebrityAndGetId(T celebrity);  // добавить Знаменитость и получить его Id
    bool UpdCelebrity(int id, T celebrity); // изменить Знаменитость по Id
    int GetCelebrityIdByName(string name);  // получить первый Id по вхождению подстроки
    int SaveChanges();                      // сохранить изменения = количество изменений
}
```

3. Приложение **DP007_json_test**, тестирует библиотеку **DP007_json.lib**. Код функции **Main** и результаты теста приведены НИЖЕ.

```
private static void Main(string[] args)
{
    Repository.JSONFileName = "Celebrities.json";
    using (ICelebrity<Celebrity> repo = Repository.Create("Celebrities")) // директорий (в текущем директории приложения)
    {
        Console.WriteLine("----- GetAllCelebrities -----");
        {
            List<Celebrity> celebrities = repo.GetAllCelebrities();
            celebrities.ForEach(c => Console.WriteLine($"Id = {c.Id}, Firstname = {c.Firstname}, Surname = {c.Surname}, PhotoPath = {c.PhotoPath} "));
        }
        Console.WriteLine("----- AddCelebrity -----");
        {
            repo.AddCelebrity(new Celebrity(0, "Marlin", "Minsky", "/Photo/Minsky.jpg"));
            List<Celebrity> celebrities = repo.GetAllCelebrities();
            celebrities.ForEach(c => Console.WriteLine($"Id = {c.Id}, Firstname = {c.Firstname}, Surname = {c.Surname}, PhotoPath = {c.PhotoPath} "));
        }
        Console.WriteLine("----- GetCelebrityIdByName, UpdCelebrity -----");
        {
            int id = repo.GetCelebrityIdByName("Marlin");
            if (id >= 0) repo.UpdCelebrity(id, new Celebrity(0, "Marvin", "Minsky", "/Photo/Minsky.jpg"));
            List<Celebrity> celebrities = repo.GetAllCelebrities();
            celebrities.ForEach(c => Console.WriteLine($"Id = {c.Id}, Firstname = {c.Firstname}, Surname = {c.Surname}, PhotoPath = {c.PhotoPath} "));
        }
        Console.WriteLine("----- GetCelebrityIdByName, DelCelebrity -----");
        {
            int id = repo.GetCelebrityIdByName("Minsky");
            if (id >= 0) repo.DelCelebrity(id);
            List<Celebrity> celebrities = repo.GetAllCelebrities();
            celebrities.ForEach(c => Console.WriteLine($"Id = {c.Id}, Firstname = {c.Firstname}, Surname = {c.Surname}, PhotoPath = {c.PhotoPath} "));
        }
        Console.WriteLine("----- AddCelebrityAndGetId -----");
        {
            int id = repo.AddCelebrityAndGetId(new Celebrity(0, "Marvin", "Minsky", "/Photo/Minsky.jpg"));
            Console.WriteLine("Minsky Id = {0}", id);
            List<Celebrity> celebrities = repo.GetAllCelebrities();
            celebrities.ForEach(c => Console.WriteLine($"Id = {c.Id}, Firstname = {c.Firstname}, Surname = {c.Surname}, PhotoPath = {c.PhotoPath} "));
        }
        {
            Console.WriteLine("SaveChanges = {0}", repo.SaveChanges());
        }
    }
    using (ICelebrity<Celebrity> repo = Repository.Create("Celebrities"))
    {
        List<Celebrity> celebrities = repo.GetAllCelebrities();
        celebrities.ForEach(c => Console.WriteLine($"Id = {c.Id}, Firstname = {c.Firstname}, Surname = {c.Surname}, PhotoPath = {c.PhotoPath} "));
    }
}
```

Консоль отладки Microsoft Visual Studio

```
----- GetAllCelebrities -----
Id = 1, Firstname = Noam, Surname = Chomsky, PhotoPath = /Photo/Chomsky.jpg
Id = 2, Firstname = Tim, Surname = Berners-Lee, PhotoPath = /Photo/Berners-Lee.jpg
Id = 3, Firstname = Edgar, Surname = Codd, PhotoPath = /Photo/Codd.jpg
Id = 4, Firstname = Donald, Surname = Knuth, PhotoPath = /Photo/Knuth.jpg
Id = 5, Firstname = Linus, Surname = Torvalds, PhotoPath = /Photo/Torvalds.jpg
Id = 6, Firstname = John, Surname = Neumann, PhotoPath = /Photo/Neumann.jpg
Id = 7, Firstname = Edsger, Surname = Dijkstra, PhotoPath = /Photo/Dijkstra.jpg
Id = 8, Firstname = Marvin, Surname = Minsky, PhotoPath = /Photo/Minsky.jpg
----- AddCelebrity -----
Id = 1, Firstname = Noam, Surname = Chomsky, PhotoPath = /Photo/Chomsky.jpg
Id = 2, Firstname = Tim, Surname = Berners-Lee, PhotoPath = /Photo/Berners-Lee.jpg
Id = 3, Firstname = Edgar, Surname = Codd, PhotoPath = /Photo/Codd.jpg
Id = 4, Firstname = Donald, Surname = Knuth, PhotoPath = /Photo/Knuth.jpg
Id = 5, Firstname = Linus, Surname = Torvalds, PhotoPath = /Photo/Torvalds.jpg
Id = 6, Firstname = John, Surname = Neumann, PhotoPath = /Photo/Neumann.jpg
Id = 7, Firstname = Edsger, Surname = Dijkstra, PhotoPath = /Photo/Dijkstra.jpg
Id = 8, Firstname = Marvin, Surname = Minsky, PhotoPath = /Photo/Minsky.jpg
Id = 9, Firstname = Marvin, Surname = Minsky, PhotoPath = /Photo/Minsky.jpg
----- GetCelebrityIdByName, UpdCelebrity -----
Id = 1, Firstname = Noam, Surname = Chomsky, PhotoPath = /Photo/Chomsky.jpg
Id = 2, Firstname = Tim, Surname = Berners-Lee, PhotoPath = /Photo/Berners-Lee.jpg
Id = 3, Firstname = Edgar, Surname = Codd, PhotoPath = /Photo/Codd.jpg
Id = 4, Firstname = Donald, Surname = Knuth, PhotoPath = /Photo/Knuth.jpg
Id = 5, Firstname = Linus, Surname = Torvalds, PhotoPath = /Photo/Torvalds.jpg
Id = 6, Firstname = John, Surname = Neumann, PhotoPath = /Photo/Neumann.jpg
Id = 7, Firstname = Edsger, Surname = Dijkstra, PhotoPath = /Photo/Dijkstra.jpg
Id = 8, Firstname = Marvin, Surname = Minsky, PhotoPath = /Photo/Minsky.jpg
Id = 9, Firstname = Marvin, Surname = Minsky, PhotoPath = /Photo/Minsky.jpg
----- GetCelebrityIdByName, DelCelebrity -----
Id = 1, Firstname = Noam, Surname = Chomsky, PhotoPath = /Photo/Chomsky.jpg
Id = 2, Firstname = Tim, Surname = Berners-Lee, PhotoPath = /Photo/Berners-Lee.jpg
Id = 3, Firstname = Edgar, Surname = Codd, PhotoPath = /Photo/Codd.jpg
Id = 4, Firstname = Donald, Surname = Knuth, PhotoPath = /Photo/Knuth.jpg
Id = 5, Firstname = Linus, Surname = Torvalds, PhotoPath = /Photo/Torvalds.jpg
Id = 6, Firstname = John, Surname = Neumann, PhotoPath = /Photo/Neumann.jpg
Id = 7, Firstname = Edsger, Surname = Dijkstra, PhotoPath = /Photo/Dijkstra.jpg
Id = 9, Firstname = Marvin, Surname = Minsky, PhotoPath = /Photo/Minsky.jpg
----- AddCelebrityAndGetId -----
```

4. Разработайте библиотеку ***DP007_ICelebrity.lib*** содержащую интерфейс ***ICelebrity***.
 5. Разработайте библиотеку ***DP007_json.lib***, применяющую библиотеку ***DP007_ICelebrity.lib***, содержащую класс ***Repository***, реализующий интерфейс ***ICelebrity***.
 6. Разработайте приложение ***DP007_json_test***.
 7. Выполните тест и обеспечьте его совпадение с результатом его выполнения, приведенным в п.3.
- 8. КОНЕЦ**